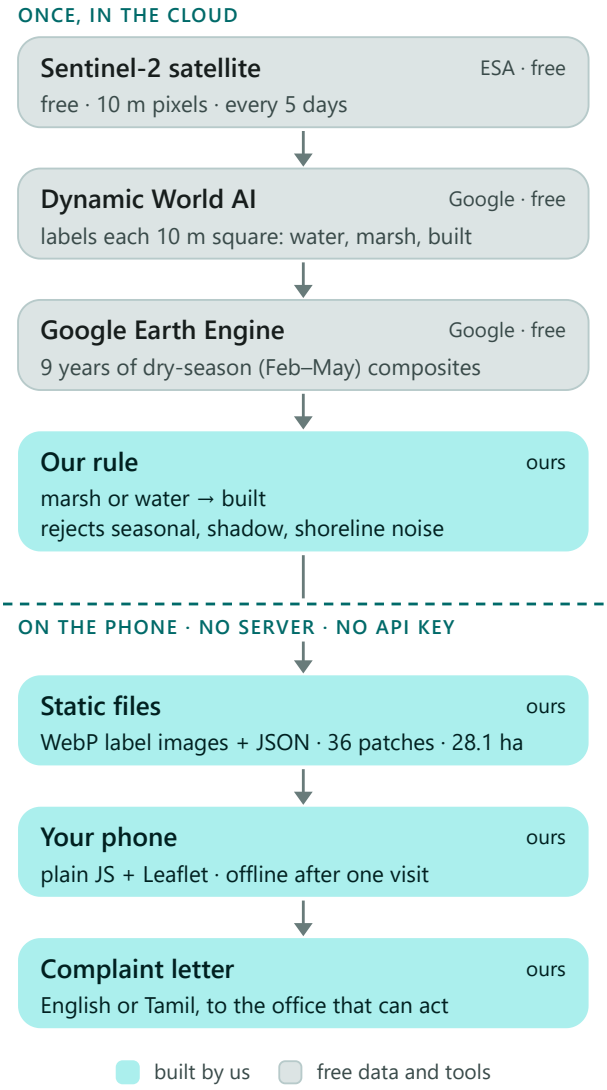
NEERKATTI

An AI water-keeper for Chennai's wetlands · Team Sternritters · DEXATHON 2026

28.1 ha	36	5	62
of marsh and water lost	patches flagged	live alerts	automated tests



WHAT IT DOES

Finds where Pallikaranai's marsh and water became buildings, season by season since 2018, from free Sentinel-2 images and Google's Dynamic World AI. Flags fresh filling as alerts, and writes a complaint letter in English or Tamil to the office that can act.

WHO THE LETTER GOES TO

- Tamil Nadu State Wetland Authority
- Greater Chennai Corporation
- Tamil Nadu Forest Department

WHY IT COSTS ₹0

The heavy analysis runs once in Google Earth Engine and is saved as plain files. The site is HTML, CSS and JavaScript: no server, no database, no API key. It works offline after one visit.

LIMITS, SAID PLAINLY

- A 10 m pixel cannot see one house; small dumping is missed.
- Dynamic World is right about buildings 86.7% of the time but catches only 36.8% of marsh, so 28.1 ha is a floor, not a ceiling.
- Every patch is a possible change that needs a ground check. NEERKATTI never names an owner.
- The photo check and the radar layer are not built yet.

Try it live
athulyapandian24.github.io/neerkatti/app/?final=1
Open on any phone. No sign-in.

62 automated tests run on every push · data: Copernicus Sentinel-2, Dynamic World (Brown et al. 2022), OpenStreetMap · code: github.com/ATHULYAPANDIAN24/neerkatti